

Battery boost (TPS61288) failure — diagnostics summary

For: an engineer picking this up cold. **Scope:** what was tested and what it rules out. Full detail in [boost-bringup-debug.md](#) — that document wins on any conflict.

The fault in one line

A *known-good* TPS61288 boost regulates 17.5 V fine **standalone** on the BT pad, then dies the instant it **drives VBUS** — **VBT**↔GND short (VIN–SW–VOUT fused) every time. **Four** battery-side boosts have failed this way; the FC channel never fails. Leading cause: a **PCB-layout asymmetry** (BT output-cap hot loop ~6× larger than FC's) — not a part, wiring, supply, or firmware problem.

Tests performed

Failure events (each destroyed a BT boost)

#	Setup	Action	Result
Death 1	9 V batteries; FC already off VBUS	Enabled BT_BUS_ENABLE (BT boost → VBUS)	Fried. Was the <i>original, never-reworked</i> boost.
Death 2	DC supply, 120 mA limit, BT only	Old FW: bus switches, then boosts	Collapsed to CC @ 120 mA; fried.
Death 3	DC supply, full ≥5 A , BT only	Booted to Idle OK; G bring-up	Hit 5–7 A, collapsed; fried.
Death 4	DC supply 8.2 V/200 mA; known-good FC boost moved to BT pad ; RC-BT matched to FC	G (gentle, bus pre-charged ~7.7 V)	Pegged 200 mA CC; fried.

Death 4 is decisive: the FC channel's *proven-good* part lived on the FC pad, then died on the BT pad → **pad/channel, not part**. Constant across all four: **VBT**↔GND short post-mortem; **BT_SEQUENCE_ENABLE** ON; 470 µF bulk cap (on V-MOT, behind **MOT_PWR_ENABLE**) OFF.

Non-destructive tests (all PASS — isolate where the fault is *not*)

Test	Result
Boost-removed bus path (DC supply on VBUS-BT , assert BT_BUS_ENABLE)	VBUS → 17.5 V, ~0 A . BT bus path is clean.
FC boost drives the bus (12 V on FC input, G)	Clean 17.5 V. Healthy reference; rig + sequence sound.
Dual-source OR-ing (live FC bus + DC on VBUS-BT , boost removed)	Bus tracked higher source smoothly, no fight. D-BT-EN passes a live bus cleanly.

Test	Result
Schematic BOM compare (<code>Scale_Car_Board_20260624.sch</code>)	Boost stages identical part-for-part. Only R1 differs — the input-V ADC divider, not a boost part.
Operator copper measurement	BT Cout sits 240 mil from the IC output pin vs FC's 40 mil (6×) ; SW pour 1.6× longer (minor).
Ohmmeter, unpowered	D-BT-EN/D-FC-EN EXP→GND and VOUT→GND all open. No static short.

Causes RULED OUT (with evidence)

- **Defective/wrong part, FB-droop, inductor** — both boosts regulate 17.5 V standalone; Death 4 was a known-good part.
- **Supply collapse / inrush / motorboating** — a stiff ≥ 5 A supply killed it as fast as 120 mA; VBUS is only ~ 40 μ F (the 470 μ F is elsewhere, and was off).
- **Static shorts** (**D-BT-EN** EXP/VOUT→GND, VBUS→GND) — ohmmeter open; FC held VBUS at 17.5 V in Death 1.
- **Dynamic low-Z fault in the BT→VBUS path** — boost-removed path ran at ~ 0 A.
- **Charger / BT_SEQUENCE static load** — path test ~ 0 A. *Caveat:* **BT_SEQUENCE** never isolated OFF; low-probability, not 100% closed.
- **Input current limit as a safety net** — Death 2 fried it at **120 mA**; destructive energy is internal ($\frac{1}{2} \cdot L \cdot di^2$), unbounded by input limit.
- **RC-BT compensation value** — REFUTED by Death 4 (matched to FC, still died).
- **0 V hot-plug / bring-up sequence** — REFUTED; Death 4's gentle **G** into a pre-charged bus still died.
- **Rework/desolder damage** — REFUTED; the Death 4 part had just worked on the FC pad.
- **Old "BT VSW 2.2× longer" trace-length figure** — SUPERSEDED; that parse excluded the polygon pours. SW is a wide low-inductance pour (1.6×, minor); the dominant metric is Cout distance.

Leading cause (strong, pending scope)

BT output-capacitor hot loop. Schematic proven symmetric → the difference is layout. All power nets are polygon pours, so trace length is irrelevant; the metric that matters is **Cout distance to the IC output pin: FC = 40 mil, BT = 240 mil (6×)**. The larger BT hot loop has higher inductance → at the di/dt of driving the bus, SW/VOUT rings past the **20 V abs-max** (only ~ 0.5 – 1.5 V of headroom over 17.5 V) → sync-rect FET fails short (**VIN-SW-VOUT**→GND, the observed post-mortem). Fits every datapoint: Death 1 (original copper), FC always survives, **RC** swap didn't help, DC injection clean (no di/dt), death at 120 mA (internal energy).

Firmware cannot fix this. Board-level remedy: add an output ceramic **right at the IC VOUT/PGND** to collapse the BT hot loop to FC-like ($\pm$ an SW snubber), or respin the BT channel.

Before the next boost: logic on a separate stiff supply; scope **VSW-BT/VOUT-BT** from $t=0$ (single-shot armed past ~ 18 V, fast kill ready). Expect a >20 V ring on BT, clean on FC. If BT doesn't ring past 20 V, re-open (ground-return/via or manufacturing defect). Four boosts have died without one scope capture — don't spend a fifth blind.